
Convergence of multi-agent systems controlled by iterative learning strategies with continuous data losses

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Abstract

To address the consistency tracking issue of multi-agent systems operating repetitive tasks with continuous data losses, this paper proposes an iterative learning control (ILC) strategy with data compensation. The convergence of multi-agent ILC systems is demonstrated by analyzing changes in the element values of the transition matrix of input errors. It is worth noting that the analysis also reveals two useful conclusions. One is that the convergence speed decreases as the probability of continuous data loss increases. The other is that the selection of the controller's learning gain depends on the maximum value of the number of continuous data losses. Simulation results verify the effectiveness of the proposed ILC strategy and the theoretical derivations regarding convergence speed and gain selection.

Keywords

Multi-agent systems; iterative learning control; continuous data loss; convergence

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Introduction

Backgrounds

There has been a growing interest in researching multi-agent systems (MASs) in recent years. Unlike single-agent systems, MASs collaborate to accomplish large-scale and complex tasks through information sharing, negotiation-based decision-making, and labor division Wang et al. (2022a). Furthermore, MASs possess remarkable self-organizing, learning, and reasoning abilities, enabling them to seek optimal solutions. As a result, MASs have found wide applications in diverse fields including intelligent transportation systems, smart grids, and multi-robot systems Shou et al. (2022); Zheng and Hou (2022); Balachennaiah and Babu (2024); Huang et al. (2019b).

By continuously adjusting the system input using previous tracking errors, iterative learning control (ILC) is an effective solution to the trajectory tracking problem of systems in repetitive tasks Qiu et al. (2020). Compared with traditional control strategies, ILC not only effectively addresses nonlinear and strongly coupled dynamic systems but also enables autonomous learning and adjustment in uncertain systems. Therefore, ILC has been extensively studied in various aspects such as initial value problems, learning algorithms, and robustness Lan et al. (2022); Nicoletti et al. (2020); Su (2021); Liu and Chi (2021).

For MASs utilizing ILC strategies, the iterative learning process of each agent relies not only on its own output errors but also on the output errors received from all other agents through wireless channels. However, the unreliability of channels presents new challenges. Due to reasons such as network link saturation, data packet disorder, noise interference and deliberate attacks, the output error transmitted through wireless channels may be lost continuously. As a result, it is necessary to study the impact of continuous data losses on the convergence of multi-agent ILC systems.

Related works

For networked iterative learning control (NILC) systems with data losses, researchers typically employ Bernoulli variables for modeling purposes. For example, Liang et al. employed Bernoulli variables to model output data losses in nonlinear non-affine systems adopting ILC strategies based on data-driven Liang et al. (2019). Chen et al. utilized Bernoulli variables to describe input and output data losses in networked systems with iterative learning model-free control strategies Chen et al. (2020). The data losses at both the measurement and actuator sides were modeled by using a Bernoulli variable for analyzing the convergence of NILC systems Huang et al. (2022). Besides Bernoulli variables, the Markov chain is another typical modeling method. Shen et al. adopted Markov chains to model data losses at both the measurement and actuator sides of linear stochastic systems based on ILC strategies Shen and Xu (2016). Wang et al. employed Markov chains to describe input and output data losses in a two-dimensional fuzzy switching system with ILC strategies Wang and Ji (2019). Herve et al. utilized Markov chains to model data losses in NILC systems Hervé et al. (2022). In addition, researchers have also utilized these two schemes to model data loss constraints in MASs.

For instance, Ni et al. used Bernoulli variables to model data losses in high-order MASs with directed network topology [Ni et al. \(2021\)](#). Shi et al. employed Bernoulli variables to model data loss issues on each communication link in discrete second-order MASs [Shi et al. \(2020\)](#). Yan et al. employed Bernoulli variables to model data loss issues in nonlinear networked MASs based on the T-S framework [Yan et al. \(2023\)](#). Xu et al. utilized Markov chains to model data loss issues under different scenarios in the channel transmission process of MASs [Xu et al. \(2019\)](#). Chen et al. used Markov chains to model the same and different data losses in discrete linear MASs [Chen et al. \(2021\)](#). Stoica et al. employed standard homogeneous Markov chains to model data loss issues in the communication channels of MASs [Stoica and Stoicu \(2022\)](#).

For NILC systems with data losses, iteration-domain and time-domain compensation have become important solutions. The former compensates for the lost data by using the received data from the corresponding moment in the previous iteration, while the latter method utilizes the data received from the previous moment within the same iteration to replace the lost data. For instance, an ILC controller with an iteration-domain compensation scheme was designed to deal with the data losses on both sides of NILC systems [Liu and Ruan \(2017\)](#). Bu et al. designed a new ILC controller using time-domain compensation to ensure system convergence [Bu et al. \(2013b\)](#). Huang et al. analyzed the data loss issues in the transmission process of NILC systems from the perspectives of time-domain and iteration-domain compensations [Huang et al. \(2019a\)](#).

For the issue of data losses during the transmission process in MASs, predictive compensation has emerged as a primary solution method. This method compensates for data losses by estimating the system state or output rather than using previous data for compensation [Pang et al. \(2021\)](#). For example, Adilla et al. proposes a new method that combines predictive strategy with a constant control gain based on the gain-to-error ratio to investigate the problem of large consecutive data loss in MASs [Adilla and Liu \(2016\)](#). Liu et al. designed a proportional predictive control strategy to solve the issue of data losses between adjacent agents in MASs [Liu \(2019\)](#). A type of increment network predictive model was proposed to actively compensate for the data losses by agents in the feedback channel [Pang et al. \(2020\)](#). Yang et al. proposed an event-triggered proactive model predictive scheme to address data losses in nonlinear MASs [Yang et al. \(2021\)](#). Cao et al. introduced a distributed predictive formation control protocol to address data losses in MASs under a leader-follower topology framework [Cao et al. \(2023\)](#). Han et al. proposed an adaptive compensation algorithm to tackle the issue of data losses in nonlinear MASs with directed topology [Han et al. \(2023\)](#).

To ensure the convergence of MASs with data losses, the ILC strategy has become a commonly used solution. For instance, Zhang et al. proposed a quantized distributed ILC strategies and introduced a decreasing learning gain sequence to address data loss problems in MASs [Zhang and Li \(2017\)](#). Deng et al. employed the ILC strategies to design a distributed control protocol to tackle data loss problems in nonlinear MASs [Deng et al. \(2019\)](#). Wu et al. proposed a predictive compensation distributed iterative learning protocol to address data loss problems in heterogeneous MASs [Wu et al. \(2020\)](#). An iterative learning controller is designed using the average-ILC strategies to handle data loss issues between adjacent agents in MASs [Li et al. \(2021\)](#). Wang et al.

proposes a model-free adaptive ILC strategy based on iterative compensation to tackle the collaborative control problem in MASs with asynchronous data losses Wang et al. (2022b).

It should be noted that although existing research has proposed various methods to ensure the consistency tracking performance of MASs under the impacts of data losses, there has been no analysis of the convergence speed of MASs. Additionally, the constraints of data losses on the learning gain of multi-agent ILC systems have not been considered.

Our contribution

Inspired by the shortcomings of existing research, this paper proposes an iterative learning strategy to ensure the convergence of multi-agent ILC systems with continuous data losses. What's more, the impacts of continuous data losses on the system convergence speed and the selection of learning gains are studied by analyzing the changes of element values in the transition matrix of tracking errors. Additionally, the stability of MASs controlled by the proposed strategy is analyzed. In summary, compared to existing achievements, the contributions of this paper mainly focus on three aspects:

- an iterative learning strategy with data compensation is proposed to ensure the convergence of multi-agent ILC systems with continuous data losses;
- it is pointed out that the convergence speed of systems decreases as the probability of continuous data loss increases;
- it is revealed that the selection of controller's learning gain depends on the maximum value of the number of continuous data losses.

In the paper, I_N represents an N -th order identity matrix. $\otimes$ indicates the Kronecker product. The notation $diag\{\cdot\}$ depicts a diagonal matrix. $E\{\cdot\}$ means the expectation factor.

Preliminaries and problem formulation

Preliminaries

The network topology among multiple agents can be represented using an undirected weighted graph $G = (V, E, A)$. In this representation, $V \in \{v_1, v_2, \dots, v_N\}$ denotes the graph's nodes and each node in set V responds an agent, The edges between the nodes are indicated by the set $E \subseteq V \times V$, and the adjacency matrix of the graph is represented by A . To specify a connection between nodes m and n , this article uses the symbol $(m, n) \in E$, which indicates that agent n can receive data from agent m . The weighted adjacency matrix A includes to diagonal elements denoted by $a_{m,n}$, where $a_{m,n} = a_{n,m} > 0$ for connected nodes m and n , and $a_{m,n} = 0$ otherwise. The set $N_m = \{m \in V | (m, n) \in E\}$ is defined as the set of neighboring nodes for the m -th agent. In the context of the directed graph G , the Laplacian matrix L is defined as the different between the matrix D and the adjacency matrix A , where $D = diag(d_1, d_1, \dots, d_N)$, and $d_m = \sum_{n \in N_m} a_{m,n}$ represents the out-degree of node m .

To ensure that MASs achieve consistency tracking of expected trajectories, a common method is to regard the desired trajectory as a virtual leader agent with the number 0, while the all agents in MASs are considered as follower agents denoted by k , where $k \in [1, 2, \dots, N]$. Subsequently, the communication topology between multiple agents can be described using a graph $\bar{G} = (v \cup \{0\}, \bar{E}, \bar{A})$ is utilized, where v represents follower agents, $\bar{E}$ indicates new boundaries, and $\bar{A}$ represents a new adjacency matrix. Moreover, the relationship between agent k and agent 0 is denoted as s_k , where $s_k > 0$ indicates a direct connection between agent k and agent 0, and $s_k = 0$ represents that the follower agent k has no connection with agent 0.

Problem formulation

Consider MASs composed of N agents, where the dynamic equation of the j -th agent is represented as:

$$\begin{cases} x_{k,i}(j+1) = Ax_{k,i}(j) + Bu_{k,i}(j) \\ y_{k,i}(j) = Cx_{k,i}(j) \end{cases} \quad (1)$$

where i indicates the iteration number. $k(k = 1, 2, \dots, N)$ means the k -th agent. $j \in [0, T]$ denotes the time. $x_{k,i}(j) \in R^m$ denotes the state of k -th agent. $u_{k,i}(j) \in R^r$ denotes the control input of j -th agent. $y_{k,i}(j) \in R^n$ denotes the output of k -th agent. A , B , and C are the suitable matrices.

Similarly, the desired trajectory is seen as a virtual leader agent and given as follows:

$$\begin{cases} x_d(j+1) = Ax_d(j) + Bu_d(j) \\ y_d(j) = Cx_d(j) \end{cases} \quad (2)$$

where $x_d(j)$, $u_d(j)$, and $y_d(j)$ denote the desired state, control input, and output, respectively.

Remark 1. To the leading-follower MASs, continuous data losses not only occurs between followers but also between the leader and followers. The reason is that only some followers can have direct access to the trajectory information from the virtual leader, and then the other followers rely on a communication topology to exchange information [Liao et al. \(2020\)](#).

ILC strategy design

The consistency error of MASs encompasses two aspects: the connection strength and the relative error. The former represents connectivity degrees between an agent and its neighbors, thereby determining the topology structure of MASs. The latter refers to the output difference between agents. In this paper, s_k denotes the connection strength between the virtual leader and follower k , $a_{k,l}$ indicates the connection strength between followers k and l , $y_d(j) - y_{k,i}(j)$ is the relative output error between the virtual leader and the follower k , and $y_{k,i}(j) - y_{l,i}(j)$ represents the relative error between followers k and l . Additionally, since there may be multiple followers directly connect to the follower k , the sum of relative errors for the follower k is utilized. Therefore, the transmitted

consistency error $\xi_{k,i}^t(j)$ is defined as follows:

$$\xi_{k,i}^t(j) = \sum_{l \in N_k} a_{k,l}(t)(y_{l,i}(j) - y_{k,i}(j)) + s_k(j)(y_d(j) - y_{k,i}(j)) \quad (3)$$

It is clear that the consistency error would be lost continuously in the time or iteration domain. In this paper, the effects of continuous data losses is described by a Bernoulli distributed variable $\eta_i(j)$. Compensating for data losses in MASs controlled by ILC strategies through networks is natural and can be implemented in either the iteration or time domain. Moreover, since the input of multi-agent ILC systems converges along the iterative axis, an iterative learning strategy based on iteration compensation is proposed as follows:

$$\begin{aligned} u_{k,i+1}(j) &= u_{k,i}(j) + \Gamma \xi_{k,i}^r(j+1) \\ &= u_{k,i}(j) + \Gamma(\eta_{k,i}(j+1)\xi_{k,i}^t(j+1) + (1 - \eta_{k,i}(j+1))\xi_{k,i-1}^r(j+1)) \end{aligned} \quad (4)$$

where Γ indicates learning gains, $\xi_{k,i}^r(j)$ represents the received consistency error.

To provide a detailed explain of the data compensation in the designed ILC strategy, equation (4) can be rewritten as:

$$\begin{aligned} u_{k,i+1}(j) &= u_{k,i}(j) + \Gamma(\eta_{k,i}(j+1)\xi_{k,i}^t(j+1) \\ &\quad + (1 - \eta_{k,i}(j+1))\eta_{k,i-1}(j+1)\xi_{k,i}^t(j+1) + \dots \\ &\quad + (1 - \eta_{k,i}(j+1))(1 - \eta_{k,i-1}(j+1)) \dots \\ &\quad \times ((1 - \eta_{k,i-d+1}(j+1))\xi_{k,i-d+1}^t(j+1) + \dots) \end{aligned} \quad (5)$$

For the sake to research the convergence of MASs controlled by the proposed ILC strategy, we make some assumptions as follows:

Assumption 1. $\eta_{k,i}(j)$ is with probabilities $\text{prob}\{\eta_{k,i}(j) = 1\} = \alpha_{k,i}$, $0 < \alpha_{k,i} < 1$, where $\alpha_{k,i}$ is known.

Assumption 2. The initial state for each iteration of MASs is the same, such as $x_{k,i}(0) = x_d(0)$, where $x_d(0)$ indicates the desired initial state of all followers.

Convergence analysis

In this section, we analyze the convergence of MASs controlled by the proposed strategy in three parts. Firstly, to simplify the analysis process, we rewrite the MASs controlled by the proposed strategy in matrix form. Secondly, we assume that only the consistency error at the $j+1$ time instant is continuously lost in the i -th and $i-1$ -th iterations and compensated for using data at the same time instant in the $i-2$ -th iteration. Finally, a general expression for d -step continuous data losses is derived.

Theorem 1. For multi-agent ILC systems with continuous data losses, if condition $0 < \|I_m \otimes \Gamma C B\| < 1$ is met, we have

$$\lim_{i \rightarrow \infty} \|\xi_i^r(j+1)\| = 0 \quad (6)$$

Proof. To analyze the convergence of multi-agent ILC systems, we transform the multi-agent ILC system in a concise form by defining

$$\begin{aligned}
 x_i(j) &= [x_{1,i}^T(j), x_{2,i}^T(j), \dots, x_{N,i}^T(j)]^T, \\
 u_i(j) &= [u_{1,i}^T(j), u_{2,i}^T(j), \dots, u_{N,i}^T(j)]^T, \\
 y_i(j) &= [y_{1,i}^T(j), y_{2,i}^T(j), \dots, y_{N,i}^T(j)]^T, \\
 \eta_i(j) &= [\eta_{1,i}^T(j), \eta_{2,i}^T(j), \dots, \eta_{N,i}^T(j)]^T, \\
 \alpha_i &= \text{diag}\{\alpha_{1,i}^T, \alpha_{2,i}^T, \dots, \alpha_{N,i}^T\}, \\
 \xi_i^t(j) &= [\xi_{1,i}^{t,T}(j), \xi_{2,i}^{t,T}(j), \dots, \xi_{N,i}^{t,T}(j)]^T, \\
 \xi_i^r(j) &= [\xi_{1,i}^{r,T}(j), \xi_{2,i}^{r,T}(j), \dots, \xi_{N,i}^{r,T}(j)]^T.
 \end{aligned} \tag{7}$$

Therefore, the MASs in (1) can be expressed as

$$\begin{cases} x_i(j+1) = (I_m \otimes A)x_i(j) + (I_m \otimes B)u_i(j), \\ y_i(j) = (I_m \otimes C)x_i(j) \end{cases} \tag{8}$$

Similarly, equation (2) can be expressed as

$$\begin{cases} x_d(j+1) = (I_m \otimes A)x_d(j) + (I_m \otimes B)u_d(j), \\ y_d(j) = (I_m \otimes C)x_d(j) \end{cases} \tag{9}$$

Defining the tracking error as $e_{k,i}(j) = y_d(j) - y_{k,i}(j)$, the consistency error in equation (3) can be given as follows:

$$\begin{aligned}
 \xi_{k,i}^t(j) &= \sum_{l \in N_k} a_{k,l}(j)(y_{l,i}(j) - y_{k,i}(j)) + s_k(j)(y_d(j) - y_{k,i}(j)) \\
 &= \sum_{l \in N_k} a_{k,l}(j)(y_d(j) - y_{k,i}(j)) - (y_d(j) - y_{l,i}(j)) + s_k(j)(y_d(j) - y_{k,i}(j)) \\
 &= \sum_{l \in N_k} a_{k,l}(j)(e_{k,i}(j) - e_{l,i}(j)) + s_k(j)e_{k,i}(j)
 \end{aligned} \tag{10}$$

Accordingly, the transmitted consistency error in equation (10) can be represented concisely as

$$\xi_i^t(j) = ((L + S) \otimes I_n)e_i(j) \tag{11}$$

where $S = \text{diag}\{s_1, s_2, \dots, s_N\}$ and $e_i(j) = [e_{1,i}^T(j), e_{2,i}^T(j), \dots, e_{N,i}^T(j)]^T$. Therefore, the iterative learning strategy proposed in (4) can be rewritten as

$$\begin{aligned}
 u_{i+1}(j) &= u_i(j) + (I_m \otimes \Gamma)\xi_i^r(j+1) \\
 &= u_i(j) + (I_m \otimes \Gamma)(\eta_i(j+1)\xi_i^t(j+1) + (1 - \eta_i(j+1))\xi_{i-1}^t(j+1))
 \end{aligned} \tag{12}$$

According to the transformed model, the convergence of multi-agent ILC systems can be analyzed. Furthermore, it is clear that the loss of consistency error at the $j + 1$ time instant does not affect the system's convergence at previous time instants. Therefore, the convergence of consistency errors can be analyzed from the following two parts.

Convergence analysis of $\xi_i^r(j + 1)$

The received consistency error can be derived as

$$\begin{aligned}\xi_i^r(j + 1) &= y_d(j + 1) - y_i(j + 1) \\ &= (I_m \otimes C)x_d(j + 1) - (I_m \otimes C)x_i(j + 1) \\ &= (I_m \otimes C)(x_d(j + 1) - x_i(j + 1)) \\ &= (I_m \otimes C)\delta x_i(j + 1)\end{aligned}\tag{13}$$

Combining equations (8) and (9), $\delta x_i(j + 1)$ in equation (13) can be derived as

$$\begin{aligned}\delta x_i(j + 1) &= x_d(j + 1) - x_i(j + 1) \\ &= (I_m \otimes A)x_d(j) + (I_m \otimes B)u_d(j) - (I_m \otimes A)x_i(j) - (I_m \otimes B)u_i(j) \\ &= (I_m \otimes A)(x_d(j) - x_i(j)) + (I_m \otimes B)(u_d(j) - u_i(j)) \\ &= (I_m \otimes A)\delta x_i(j) + (I_m \otimes B)\delta u_i(j) \\ &= (I_m \otimes A)^{j+1}\delta x_i(0) + \sum_{h=0}^j (I_m \otimes A)^{j-h}(I_m \otimes B)\delta u_i(h) \\ &= (I_m \otimes A)^{j+1}\delta x_i(0) + \sum_{h=0}^j (I_m \otimes A^{j-h}B)\delta u_i(h)\end{aligned}\tag{14}$$

If $\xi_i^t(j + 1)$ is lost, $\xi_i^r(j + 1) = \xi_{i-1}^r(j + 1)$. Therefore, $\delta u_{i+1}(j)$ can be represented as

$$\begin{aligned}\delta u_{i+1}(j) &= u_d(j) - u_i(j) - [\eta_i(j + 1)\xi_i^r(j + 1) \\ &\quad + (1 - \eta_i(j + 1))\xi_{i-1}^r(j + 1)] \otimes \Gamma \\ &= \delta u_i(j) - (I_m \otimes \Gamma C)[\eta_i(j + 1)\delta x_i(j + 1) \\ &\quad + (1 - \eta_i(j + 1))\delta x_{i-1}(j + 1)] \\ &= \delta u_i(j) - \sum_{h=0}^j (I_m \otimes \Gamma C A^{j-h}B) \\ &\quad \times [\eta_i(h + 1)\delta u_i(h) + (1 - \eta_i(h + 1))\delta u_{i-1}(h)]\end{aligned}\tag{15}$$

According to equation (15), it can be seen that the consistency error $\xi_i^r(j + 1)$ is compensated by $\xi_{i-1}^r(j + 1)$. Defining $M^b = (I_m \otimes \Gamma C A^b B)$, $b \in (0, \dots, j)$,

$\delta u_{i+1}(j)$ can be rewritten as

$$\begin{aligned}\delta u_{i+1}(j) &= \delta u_i(j) - (I_m \otimes \Gamma C B) \delta u_{i-1}(j) - \sum_{h=0}^{j-1} (I_m \otimes \Gamma C A^{j-h} B) \delta u_i(h) \\ &= G_{i+1|i}(j) \cdot \varphi_i(j) - (I_m \otimes \Gamma C B) \delta u_{i-1}(j)\end{aligned}\quad (16)$$

where

$$G_{i+1|i}(j) = \begin{bmatrix} -M^j & \cdots & -M^1 & I \end{bmatrix} \quad (17)$$

$$\varphi_i(j) = \begin{bmatrix} \delta u_i(0) & \delta u_i(1) & \cdots & \delta u_i(j) \end{bmatrix}^T \quad (18)$$

When the consistency error $\xi_{i-1}^r(j+1)$ is lost continuously and compensated by $\xi_{i-2}^r(j+1)$, $\delta u_{i-1}(j) = \delta u_{i-2}(j)$. As a result, we can obtain

$$\delta u_i(b) = \delta u_{i-1}(b) - (I_m \otimes \Gamma C B) \delta u_{i-1}(b) - \sum_{b=0}^{j-1} (I_m \otimes \Gamma C A^{j-b} B) \delta u_{i-1}(b) \quad (19)$$

$$\delta u_i(j) = \delta u_{i-1}(j) - (I_m \otimes \Gamma C B) \delta u_{i-2}(j) - \sum_{h=0}^{j-1} (I_m \otimes \Gamma C A^{j-h} B) \delta u_{i-1}(h) \quad (20)$$

Equations (19) and (20) can be rewritten in a compact form as

$$\varphi_i(j) = G_{i,i-1}(j) \cdot \varphi_{i-1}(j) + \begin{bmatrix} 0 & \cdots & 0 & -M^0 \end{bmatrix} \delta u_{i-2}(j) \quad (21)$$

where

$$G_{i,i-1}(j) = \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -M^{j-1} & \cdots & -M^1 & I - M^0 & 0 \\ -M^j & \cdots & \cdots & -M^1 & I \end{bmatrix} \quad (22)$$

$$\varphi_{i-1}(j) = \begin{bmatrix} \delta u_{i-1}(0) & \delta u_{i-1}(1) & \cdots & \delta u_{i-1}(j) \end{bmatrix}^T \quad (23)$$

If consistency error $\xi_{i-2}^r(j+1)$ is successfully received, we can obtain $\xi_{i-1}^r(j+1) = \xi_{i-2}^r(j+1) = \xi_{i-2}^t(j+1)$, and $\varphi_{i-1}(j)$ can be expressed as

$$\varphi_{i-1}(j) = G_{i-1,i-2}(j) \cdot \varphi_{i-2}(j) \quad (24)$$

where

$$G_{i-1,i-2}(j) = \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -M^{j-1} & \cdots & -M^1 & I - M^0 & 0 \\ -M^j & \cdots & \cdots & -M^1 & I - 3M^0 \end{bmatrix} \quad (25)$$

$$\varphi_{i-2}(j) = [\delta u_{i-2}(0) \quad \delta u_{i-2}(1) \quad \cdots \quad \delta u_{i-2}(j)]^T \quad (26)$$

Consequently, combining equations (16), (21) and (24), $\delta u_{i+1}(j)$ can be rewritten as

$$\delta u_{i+1}(j) = G_{i+1,i}(j) \cdot G_{i,i-1}(j) \cdot G_{i-1,i-2}(j) \cdot \varphi_{i-2}(j) \quad (27)$$

So far, the transition matrix of input errors is derived under the effects of two-step continuous data losses. By analogy, it can be inferred that when $\xi_i^r(j+1), \xi_{i-1}^r(j+1), \dots, \xi_{i+1-d}^r(j+1)$ are continuously lost and compensated by $\xi_{i-d}^r(j+1)$, where d represents the number of continuous data losses in the iteration domain, $\delta u_{i+1}(j)$ can be represented as follows:

$$\delta u_{i+1}(j) = \prod_{b=0}^{d-1} G_{i+1-b,i-b}(j) \cdot G_{i+1-d,i-d}(j) \cdot \varphi_{i-d}(j) \quad (28)$$

where

$$\begin{aligned} G_{i+1-b,i-b}(j) &= G_{i,i-1}(j) \\ &= \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -M^{j-1} & \cdots & -M^1 & I - M^0 & 0 \\ -M^j & \cdots & \cdots & -M^1 & I \end{bmatrix} \end{aligned} \quad (29)$$

$$G_{i+1-d,i-d}(j) = \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -M^{j-1} & \cdots & -M^1 & I - M^0 & 0 \\ -M^j & \cdots & \cdots & -M^1 & I - (d+1)M^0 \end{bmatrix} \quad (30)$$

According to equations (28), (29) and (30), it is easy to find that $G_{i+1-b,i-b}(j)$ and $G_{i+1-d,i-d}(j)$ are both lower triangular matrices, so the condition $0 < \|I_m \otimes \Gamma CB\| < 1$ ensures $0 < \|I - (I_m \otimes \Gamma CB)\| < 1$ and $0 < \|I - (d+1)(I_m \otimes \Gamma CB)\| < 1$. As a result, we can obtain $\lambda_s(\prod_{i=0}^{\infty} \|G_{i+1,i}(j)\|) \rightarrow 0, s = 0, \dots, j$ and deduce $\lim_{i \rightarrow \infty} \|\delta u_{i+1}(j)\| = 0$. Correspondingly, it can be proven that $\lim_{i \rightarrow \infty} \|\xi_i^r(j+1)\| = 0$.

It can be also found that some elements in the transition matrices are changed. To be specific, the elements at $(j+1, j+1)$ in $G_{i+1-b,i-b}(j)$ are all altered from $I - (I_m \otimes \Gamma CB)$ to I , while the element at $(j+1, j+1)$ in $G_{i+1-d,i-d}(j)$ is altered from $I - (I_m \otimes \Gamma CB)$ to $I - (d+1)(I_m \otimes \Gamma CB)$. Due to the elements at $(j+1, j+1)$ in $G_{i+1-b,i-b}(j)$ are all increased from $I - (I_m \otimes \Gamma CB)$ to I , the system's convergence is decelerated at time $j+1$. Consequently, the larger the value of b , the more data is

lost, and the more elements are increased from $I - (I_m \otimes \Gamma CB)$ to I . Consequently, the system's convergence speed at time $j + 1$ becomes slower.

Additionally, due to the element at $(j + 1, j + 1)$ in $G_{i+1-d|i-d}(j)$ is altered from $I - (I_m \otimes \Gamma CB)$ to $I - (d + 1)(I_m \otimes \Gamma CB)$, it can be seen that continuous data losses limit the selection of controller's learning gain. The maximum value of continuous data losses is denoted as $d_{\max} = T \cdot P_l$, where P_l represents the loss probability. When the selection of learning gain satisfies $0 < \|I_m \otimes \Gamma CB\| < \frac{2}{d_{\max} + 1}$, then the condition $0 < \|I - (d_{\max} + 1)(I_m \otimes \Gamma CB)\| < 1$ is satisfied. As a result, $0 < \|I - (d + 1)(I_m \otimes \Gamma CB)\| < 1$ is satisfied for all values of d , ensuring the convergence of the consistency error $\xi_i^r(j + 1)$.

Convergence analysis of $\xi_i^r(j + 1 + h)$

In this subsection, we continue to analyze the convergence of consistency error $\|\xi_i^r(j + 1 + h)\|$, $h = 1, \dots, T - j - 1$. Due to the similarity with the above analysis process, it is possible to analyze the convergence of $\|\xi_i^r(j + 2)\|$ and then prove the convergence of $\|\xi_i^r(j + 1 + h)\|$, $h = 1, \dots, T - j - 1$ by analogy. The consistency error $\xi_i^r(j + 2)$ can be expressed as

$$\begin{aligned} \xi_i^r(j + 2) &= y_d(j + 2) - y_i(j + 2) \\ &= \sum_{h=0}^{j+1} (I_m \otimes A^{j+1-h} BC) \delta u_i(h) \end{aligned} \quad (31)$$

Furthermore, $\delta u_{i+1}(j + 1)$ can be expressed as

$$\begin{aligned} \delta u_{i+1}(j + 1) &= u_d(j + 1) - u_i(j + 1) - [\eta_i(j + 2) \xi_i^r(j + 2) \\ &\quad + (1 - \eta_i(j + 2)) \xi_{i-1}^r(j + 2)] \otimes \Gamma \\ &= \delta u_i(j + 1) - (I_m \otimes \Gamma C) [\eta_i(j + 2) \delta x_i(j + 2) \\ &\quad + (1 - \eta_i(j + 2)) \delta x_{i-1}(j + 1)] \\ &= \delta u_i(j + 1) - \sum_{h=0}^{j+1} (I_m \otimes \Gamma C A^{j+1-h} B) \\ &\quad \times [\eta_i(h + 1) \delta u_i(h) + (1 - \eta_i(h + 1)) \delta u_{i-1}(h)] \end{aligned} \quad (32)$$

According to equation (12), it can be easy seen that the consistency error $\xi_i^r(j + 1)$ is compensated by $\xi_{i-1}^r(j + 1)$, equation (32) can be rewritten as

$$\begin{aligned} \delta u_{i+1}(j + 1) &= \delta u_i(j + 1) - (I_m \otimes \Gamma CB) \delta u_i(j + 1) \\ &\quad - (I_m \otimes \Gamma CAB) \delta u_{i-1}(j) - \sum_{h=0}^{j-1} (I_m \otimes \Gamma C A^{j+1-h} B) \delta u_i(h) \\ &= G_{i+1,i}(j + 1) \cdot \varphi_i(j + 1) - (I_m \otimes \Gamma CAB) \delta u_{i-1}(j) \end{aligned} \quad (33)$$

where

$$G_{i+1,i}(j + 1) = \begin{bmatrix} -M^{j+1} & \dots & -M^2 & 0 & I - M^0 \end{bmatrix} \quad (34)$$

$$\varphi_i(j+1) = [\delta u_i(0) \quad \delta u_i(1) \quad \cdots \quad \delta u_i(j+1)]^T \quad (35)$$

Similarly, if the consistency error $\xi_{i-1}^r(j+1)$ is lost and compensated by $\xi_{i-2}^r(j+1)$, it can be further inferred that

$$\begin{aligned} \delta u_i(j+1) &= \delta u_i(j) - (I_m \otimes \Gamma C B) \delta u_i(j+1) \\ &\quad - (I_m \otimes \Gamma C A B) \delta u_{i-2}(j) - \sum_{h=0}^{j-1} (I_m \otimes \Gamma C A^{j+1-h} B) \delta u_{i-1}(h) \end{aligned} \quad (36)$$

$$\delta u_i(b) = \delta u_{i-1}(b) - (I_m \otimes \Gamma C B) \delta u_{i-1}(b) - \sum_{b=0}^{j-1} (I_m \otimes \Gamma C A^{j+1-b} B) \delta u_{i-1}(b) \quad (37)$$

where $0 \leq b \leq j-1$. Equations (36) and (37) can be rewritten in a compact form as

$$\begin{aligned} \varphi_i(j+1) &= G_{i,i-1}(j+1) \cdot \varphi_{i-1}(j+1) \\ &\quad + [0 \quad \cdots \quad 0 \quad 1 \quad (I_m \otimes \Gamma C A B)]^T \delta u_{i-2}(j) \end{aligned} \quad (38)$$

where

$$G_{i,i-1}(j+1) = \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ -M^{j-1} & \cdots & \cdots & I - M^0 & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 & I & 0 \\ -M^{j+1} & \cdots & \cdots & -M^1 & 0 & I - M^0 \end{bmatrix} \quad (39)$$

$$\varphi_{i-1}(j+1) = [\delta u_{i-1}(0) \quad \delta u_{i-1}(1) \quad \cdots \quad \delta u_{i-1}(j+1)]^T \quad (40)$$

When the consistency error $\xi_{i-2}^r(j+1)$ is received, $\xi_{i-1}^r(j+1) = \xi_{i-2}^r(j+1) = \xi_{i-2}^t(j+1)$, and $\varphi_{i-1}(j+1)$ can be represented as

$$\varphi_{i-1}(j+1) = G_{i-1,i-2}(j+1) \cdot \varphi_{i-2}(j+1) \quad (41)$$

where

$$G_{i-1,i-2}(j+1) = \begin{bmatrix} I - M^0 & 0 & \cdots & \cdots & 0 \\ -M^1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \cdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & I & 0 \\ -M^{j+1} & \cdots & \cdots & -3M^1 & I - M^0 \end{bmatrix} \quad (42)$$

$$\varphi_{i-1}(j+1) = [\delta u_{i-2}(0) \quad \delta u_{i-2}(1) \quad \cdots \quad \delta u_{i-2}(j+1)]^T \quad (43)$$

Therefore, combining equations (32), (37) and (40), $\delta u_{i+1}(j+1)$ can be expressed using $\varphi_{i-2}(j+1)$ as

$$\varphi_{i+1}(j+1) = G_{i+1,i}(j+1) \cdot G_{i,i-1}(j+1) \cdot G_{i-1,i-2}(j+1) \cdot \varphi_{i-2}(j+1) \quad (44)$$

When $\xi_i^r(j+1), \xi_{i-1}^r(j+1), \dots, \xi_{i+1-d}^r(j+1)$ are lost continuously and all replaced by $\xi_{i-d}^r(j+1)$, $\delta u_{i+1}(j+1)$ can be derived using $\varphi_{i-d}(j+1) = [\delta u_{i-d}(0) \dots \delta u_{i-d}(j+1)]^T$ as

$$\delta u_{i+1}(j+1) = \prod_{b=0}^{d-1} G_{i+1-b,i-b}(j+1) \cdot G_{i+1-d,i-d}(j+1) \cdot \varphi_{i-d}(j+1) \quad (45)$$

where

$$\begin{aligned} G_{i+1-b,i-b}(j+1) &= H_{i,i-1}(j+1) \\ &= \begin{bmatrix} I - M^0 & 0 & \dots & \dots & \dots & 0 \\ -M^1 & \ddots & \ddots & & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ -M^{j-1} & \dots & \dots & I - M^0 & \ddots & \vdots \\ 0 & \dots & \dots & 0 & I & 0 \\ -M^{j+1} & \dots & \dots & -M^1 & 0 & I - M^0 \end{bmatrix} \end{aligned} \quad (46)$$

$$G_{i+1-d,i-d}(j+1) = \begin{bmatrix} I - M^0 & 0 & \dots & \dots & 0 \\ -M^1 & \ddots & & & \vdots \\ \vdots & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & I & 0 \\ -M^{j+1} & \dots & \dots & -(d+1)M^1 & I - M^0 \end{bmatrix} \quad (47)$$

Due to $G_{i+1-b|i-b}(j+1)$ and $G_{i+1-d|i-d}(j+1)$ are lower triangular matrices, it is easy to find that the convergence condition $0 < \|I_m \otimes \Gamma CB\| < 1$ guarantees $\lambda_s(\prod_{i=0}^{\infty} \|G_{i+1,i}(j)\|) \rightarrow 0 (s = 0, \dots, j+1)$ and then $\lim_{i \rightarrow \infty} \|\delta u_{i+1}(j+1)\| = 0$. Correspondingly, $\lim_{i \rightarrow \infty} \|\xi_i^r(j+2)\| = 0$ is proven.

As to $\|\delta u_{i+1}(j+1+h)\| (2 \leq h \leq T-j-1)$, it is easy to observe that the element values in $(h+1)$ -th row of $G_{i+1-b|i-b}(j+1)$ and $G_{i+1-d|i-d}(j+1)$ are all increased from $I - (I_m \otimes \Gamma CB)$ to I . It can easily obtain that $\lambda_s(\prod_{i=0}^{\infty} \|G_{i+1,i}(f)\|) \rightarrow 0, s = 0, \dots, f$ ($f = j+2, \dots, T-1$) and then $\lim_{i \rightarrow \infty} \|\delta u_{i+1}(h)\| = 0$ ($2 \leq h \leq T-j-1$). As a result, it can be proven that $\lim_{i \rightarrow \infty} \|\xi_i^r(j+1+h)\| = 0$ ($2 \leq h \leq T-j-1$).

Additionally, continuous data losses decreases the system's convergence speed at time $j + 1 + h, 1 \leq h \leq T - j - 1$. The reason is that the element values in $(j + 1)$ -th row of $G_{i+1-b, i-b}(j + 1 + h)$ are all increased from $I - (I_m \otimes \Gamma CB)$ to I .

This completes the proof.

Based on the analysis in the above two parts, it can be found that the condition $0 < \|I_m \otimes \Gamma CB\| < 1$ guarantees the convergence of multi-agents controlled by the proposed ILC strategy. Moreover, it can be seen that continuous data losses not only decrease the system convergence speed but also limit the selection of learning gain.

Remark 2. The ILC strategy proposed for MASs is based on fixed and undirected topologies without communication delays. It should be pointed out that the topology may be dynamic or directed. Changes in the topology will affect the strength of connections between leaders and followers, as well as among the followers themselves. Furthermore, the alteration in connection strength only affects the element values of the connection matrices L and S . With the proposed control strategy, the convergence of MASs can still be ensured. However, changes in the element values of the connection matrices L and S limits the range of controller learning gains, thereby impacting the convergence speed of MASs controlled by the proposed control strategy. Additionally, fixed or random communication delays among agents may exist. Fixed delays can be handled by appropriately advancing output errors for controller learning, while random delays can be described using Bernoulli-distributed random variables and compensated for along with data losses.

Remark 3. The increase in the scale of MASs will bring new challenges to system control and communication. For large-scale MASs, the dimension of the matrix $L + S$ composed of communication connections between agents also grows larger. Therefore, the computational complexity of learning control increases accordingly. An effective method for reducing computational complexity is to apply the proposed learning control strategy in a hierarchical manner, which divides a large-scale system into multiple smaller sub-systems [Uhruski et al. \(2008\)](#); [Wang et al. \(2020\)](#). In other words, an agent can act as both a follower in the large-scale system and a leader in a sub-system. Correspondingly, the proposed control strategy can be used hierarchically to control large-scale MASs with hundreds or thousands of agents. As for the communication overhead caused by the increase in system scale, three schemes can be employed. Firstly, quantizers can be used to reduce the amount of data shared among agents [Han et al. \(2023\)](#); [Wang et al. \(2024\)](#). Secondly, a distributed search method can be employed to dynamically select appropriate agents as leaders, thereby optimizing communication inefficiencies [Jang and Agha \(2006\)](#); [Chen et al. \(2024a\)](#). Finally, event-triggered strategies can be adopted to effectively reduce communication frequencies [Chen et al. \(2024b\)](#).

Stability analysis

In this section, we continue to analyze the stability of multi-agent controlled by the proposed strategy in the presence of continuous data losses. Firstly, the system would

be rewritten to facilitate later analysis. Then, the Lipschitz continuity condition is used to derive the system's stability.

Two definitions are given for the discussion.

Definition 1. The λ -norm for a positive real function $g(\cdot)$ is defined as Wang (1998)

$$\|g(t)\|_\lambda = \sup_{t \in [0, T]} \{\lambda^t \|g(t)\|\}, 0 < \lambda < 1 \quad (48)$$

Definition 2. Suppose that a real positive series $\{a_k\}$ satisfies Bu et al. (2013a)

$$a_k \leq p_0 a_{k-1} + p_1 a_{k-2} + \dots + p_n a_{k-n} + \varepsilon \quad (49)$$

where $p_i \geq 0$, ($i = 0, 1, \dots, n$), $\varepsilon \geq 0$, and $p = \sum_{i=0}^n p_i < 1$.

Then the following that

$$\lim_{k \rightarrow \infty} a_k \leq \frac{\varepsilon}{1-p} \quad (50)$$

Theorem 2. The MASs controlled by the proposed learning strategy, even with continuous data losses, remain stable if the consistency error $\xi_{i+1}^r(j+1)$ satisfies

$$\lim_{i \rightarrow \infty} \|\xi_i^r(j+1)\|_\lambda \leq \frac{\varepsilon}{1-p} \quad (51)$$

Proof. The consistency error $\xi_{i+1}^r(j+1)$ can be obtained as

$$\begin{aligned} \xi_{i+1}^r(j+1) &= y_d(j+1) - y_{i+1}(j+1) \\ &= \xi_i^r(j+1) + y_i(j+1) - y_{i+1}(j+1) \end{aligned} \quad (52)$$

Based on (8), the consistency error $\xi_{i+1}^r(j+1)$ can be further derived as

$$\xi_{i+1}^r(j+1) = \xi_i^r(j+1) - (I_m \otimes C)(x_{i+1}(j+1) - x_i(j+1)) \quad (53)$$

According to (8), $x_i(j+1)$ can be expressed as

$$\begin{aligned} x_i(j+1) &= (I_m \otimes A)x_i(j) + (I_m \otimes B)u_i(j) \\ &= (I_m \otimes A^{j+1})x_i(0) + \sum_{\tau=0}^j (I_m \otimes A^{j-\tau} B)u_i(\tau) \end{aligned} \quad (54)$$

Similarly, the proposed ILC strategy in (4) can be rewritten as

$$u_{i+1}(j) = u_i(j) + (I_m \otimes \Gamma) \sum_{h=0}^d \theta_h \xi_{i-h}^r(j+1) \quad (55)$$

where $\theta_0 = \alpha_i$, $\theta_h = \prod_{g=0}^{h-1} (1 - \alpha_{i-g}) \alpha_{i-h}$, $1 \leq h \leq d$, and $\theta_d = \prod_{h=0}^d (1 - \alpha_{i-h})$.

Substituting (54) into (53), we have

$$\begin{aligned}
\xi_{i+1}^r(j+1) &= \xi_i^r(j+1) - (I_m \otimes C)[(I_m \otimes A^{j+1})x_{i+1}(0) \\
&\quad + \sum_{\tau=0}^j (I_m \otimes A^{j-\tau}B)u_{i+1}(\tau) \\
&\quad - (I_m \otimes A^{j+1})x_i(0) - \sum_{\tau=0}^j (I_m \otimes A^{j-\tau}B)u_i(\tau)] \\
&= \xi_i^r(j+1) - (I_m \otimes C) \sum_{\tau=0}^j (I_m \otimes A^{j-\tau}B)(u_{i+1}(\tau) - u_i(\tau)) \\
&\quad - (I_m \otimes CA^{j+1})(x_{i+1}(0) - x_i(0))
\end{aligned} \tag{56}$$

Furthermore, substituting (55) into (56), we have

$$\begin{aligned}
\xi_{i+1}^r(j+1) &= \xi_i^r(j+1) - (I_m \otimes \Gamma C) \sum_{\tau=0}^j (I_m \otimes A^{j-\tau}B) \sum_{h=0}^d \theta_h \xi_{i-h}^r(\tau+1) \\
&\quad - (I_m \otimes CA^{j+1})(x_{i+1}(0) - x_i(0)) \\
&= (I - \theta_0(I_m \otimes \Gamma CB))\xi_i^r(j+1) - (I_m \otimes \Gamma CB) \sum_{h=1}^d \theta_h \xi_{i-h}^r(j+1) \\
&\quad - (I_m \otimes \Gamma C) \sum_{\tau=0}^{j-1} (I_m \otimes A^{j-\tau}B) \sum_{h=0}^d \theta_h \xi_{i-h}^r(\tau+1) \\
&\quad - (I_m \otimes CA^{j+1})(x_{i+1}(0) - x_i(0))
\end{aligned} \tag{57}$$

According to equation (57), we obtain

$$\begin{aligned}
&\|\xi_{i+1}^r(j+1)\| \\
&\leq \|I - \theta_0(I_m \otimes \Gamma CB)\| \|\xi_i^r(j+1)\| + \|(I_m \otimes \Gamma CB) \sum_{h=1}^d \theta_h \xi_{i-h}^r(j+1)\| \\
&\quad + \sum_{\tau=0}^{j-1} \|(I_m \otimes \Gamma CA^{j-\tau}B)\| \sum_{h=0}^d \theta_h \xi_{i-h}^r(\tau+1) \\
&\quad + \|(I_m \otimes CA^{j+1})(x_{i+1}(0) - x_i(0))\| \\
&\leq \sum_{h=0}^d p_h \|\xi_{i-h}^r(j+1)\| + \sum_{h=0}^d b_h \sum_{\tau=0}^{j-1} \|\xi_{i-h}^r(\tau+1)\| + \varepsilon
\end{aligned} \tag{58}$$

where

$$\begin{aligned} p_0 &= \|I - \theta_0(I_m \otimes \Gamma CB)\|, \\ p_h &= \|\theta_h(I_m \otimes \Gamma CB)\|, h \in [1, d], \\ b_h &= \sup_{j \in [0, T], \tau \in [0, j]} \|\theta_h(I_m \otimes \Gamma CA^{j-\tau} B)\|, h \in [1, d], \\ \varepsilon &= (I_n \otimes CA^{j+1})(x_{i+1}(0) - x_i(0)). \end{aligned}$$

Therefore, the λ norm of $\xi_{i+1}^r(j+1)$ is derived as

$$\begin{aligned} & \|\xi_{i+1}^r(j+1)\|_\lambda \\ & \leq \sum_{h=0}^d p_h \|\xi_{i-h}^r(j+1)\|_\lambda + \sum_{h=0}^d b_h \frac{\lambda(1-\lambda^T)}{1-\lambda} \|\xi_{i-h}^r(j+1)\|_\lambda + \varepsilon \\ & \leq \sum_{h=0}^d (p_h + b_h \frac{\lambda(1-\lambda^T)}{1-\lambda}) \|\xi_{i-h}^r(j+1)\|_\lambda + \varepsilon \end{aligned} \quad (59)$$

When $0 < p < 1$ and λ is sufficiently small, we can ensure

$$0 < p = \sum_{h=0}^d p_h + b_h \frac{\lambda(1-\lambda^T)}{1-\lambda} < 1. \quad (60)$$

Thus, equations (59) and (60) mean that when $j \in [0, T]$ and $\lim_{i \rightarrow \infty} \|\xi_i^r(j+1)\|_\lambda \leq \frac{\varepsilon}{1-p}$ hold, the system remains stable.

Simulation results

In the section, this paper considers MASs composed of three brush DC permanent magnet linear motors [Yu et al. \(2013\)](#) as follows:

$$\begin{cases} \dot{x}(j) = v(j) \\ v(j) = \frac{u(j) - (f_c + (f_s - f_c)e^{-(\dot{x}/\dot{x}_\delta)^\delta} + f_v \dot{x}) \operatorname{sgn} \dot{x} - b_1 \sin(\omega_0 x(t))}{m} \end{cases}$$

where $x(j)$ indicates position (m), $u(j)$ represents thrust (N), $v(j)$ denotes velocity (m/s), m represents mass (kg), j indicates continuous time (s), f_c represents the minimum value of Coulomb friction (N), f_v denotes load parameters ($N \cdot s \cdot m^{-1}$), f_s indicates static friction (N), δ denotes empirical parameters, $\dot{x}_\delta$ represents lubrication parameters, b_1 indicates amplitude (N) and ω_0 denotes angular velocity (s^{-1}).

The model parameters are as represents $\delta = 1$, $\dot{x}_\delta = 0.1$, $u(j) = 0$, $v(j) = 0$, $m = 0.59kg$, $f_c = 10N$, $f_v = 10N \cdot s \cdot m^{-1}$, $f_s = 20N$, $b_1 = 8.5N$ and $\omega_0 = 314s^{-1}$.

The expected trajectory is regarded as the virtual leader and given as follows:

$$v_d(\tau) = (x_0 - x_l)(60\tau^3 - 30\tau^4 - 30\tau^2)$$

where $\tau = j/(j_l - j_0)$, x_l and x_0 represent terminal and initial position, respectively. $x_0 = 0$, $x_l = 2m$, $j_l = 1s$. the sampling interval is 0.01.

The communication topology structure among multi-agents is shown in Figure 1. Node 0 in Figure 1 represents the virtual leader. The followers can directly obtain the expected trajectory information from the leader and can transmit information to each other. Therefore, based on graph theory knowledge, it can be inferred that $S = \text{diag}(0.5, 0.5, 0.5)$, and the Laplacian matrix is denoted as

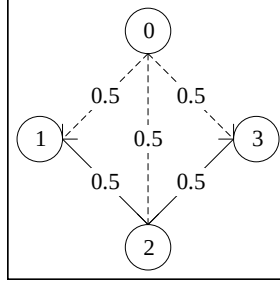


Figure 1. Communication topology structure among multi-agents

$$L = \begin{bmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 1 & -0.5 \\ 0 & -0.5 & 0.5 \end{bmatrix}$$

The loss rates are set to 0.15, 0.20, and 0.30 respectively. Correspondingly, the maximum values for continuous data losses are 15, 20, and 30, respectively. By calculation, it is known that the maximum values of the controller's learning gain are 0.8, 0.75, and 0.66, respectively. To prove the impact of the controller's learning gain selection on the convergence of the system, this section set the learning gain as $\Gamma = 0.6$, 0.7 and 0.76, respectively. It can be seen that $\Gamma = 0.6$ is supposed to ensure the system convergence under three loss rates, $\Gamma = 0.7$ is supposed to ensure the system convergence when loss rate is 0.15 and 0.20, while $\Gamma = 0.76$ would only do well to ensure the system convergence when loss rate is 0.15. The arrangement of the simulation experiments is as follows: we first verified the effectiveness of the proposed learning control strategy and the impact of data loss probability on system convergence speed. Then, we evaluated the constraints imposed by data loss probability on system learning gain selection from both convergence curves and tracking trajectories. Finally, we used the control strategy without iterative compensation from [Chen et al. \(2023\)](#) as a benchmark to assess the role of iterative compensation in enhancing system convergence speed.

Figure 2 displays the convergence and stability of consistency error mean for each agent when $\Gamma = 0.6$. Because 0.6 is smaller than the controller's maximum learning gain under three data losses, it is evident that the consistency error mean converges to 0. Furthermore, when the loss rate is higher, the convergence speed of the system is slower. Figure 3 illustrates the convergence and stability of consistency error mean for each

agent when $\Gamma = 0.7$. It is observed that the convergence and stability of consistency error mean is guaranteed when the data loss rates are 0.15 and 0.20, respectively. However, the convergence and stability of consistency error mean cannot be ensured when the data loss rate is 0.30 because the controller's learning gain 0.7 exceeds 0.66. Figure 4 represents the convergence and stability of consistency error mean for each agent when $\Gamma = 0.76$. It is evident that the controller's learning gain 0.76 is greater than 0.66 and 0.75 and less than 0.8. Therefore, the convergence and stability of consistency error mean is ensured when the data loss rate is 0.15 but not guaranteed when data loss rates are 0.20 and 0.30.

Figure 5 represents the output generated by the proposed ILC strategy when $\Gamma = 0.7$ with three data loss rates. Obviously, output trajectories at 50-th iteration can track the expected trajectory when the consistency error loss rates are 0.15 and 0.20, but the tracking performance of the output trajectories cannot be guaranteed when the loss rate is 0.30. This figure further verifies the effectiveness of the proposed ILC strategy and theoretical derivations regarding convergence speed and gain selection.

To verify that the proposed strategy with iteration domain compensation has a faster convergence speed, the control strategy without compensation from [Chen et al. \(2023\)](#) is adopted as a benchmark. Figure 6 represents the convergence of the absolute mean of agent output errors for the strategy in [Chen et al. \(2023\)](#) and the strategy proposed in this paper. It is evident that the control strategy proposed in this paper converges faster than the strategy in [Chen et al. \(2023\)](#) under the same data loss probabilities. This is because the control strategy in [Chen et al. \(2023\)](#) only utilizes system characteristics to ensure convergence and ignores a data compensation strategy to address the data loss problems.

Conclusions

In this paper, an iterative learning strategy with data compensation has been proposed to ensure the convergence of multi-agent ILC systems with continuous data losses. Based on the proposed learning strategy, we have analyzed the variation of the element values in the transition matrix of input errors. The analysis has shown that the system's convergence speed decreases as the probability of continuous data loss increases. Additionally, it is revealed that the selection of controller's learning gain depends on the maximum value of the number of continuous data losses.

In future work, we will research resilient filtering methods to address uncertainties in agent dynamics, noise, external disturbances, and parameter variations.

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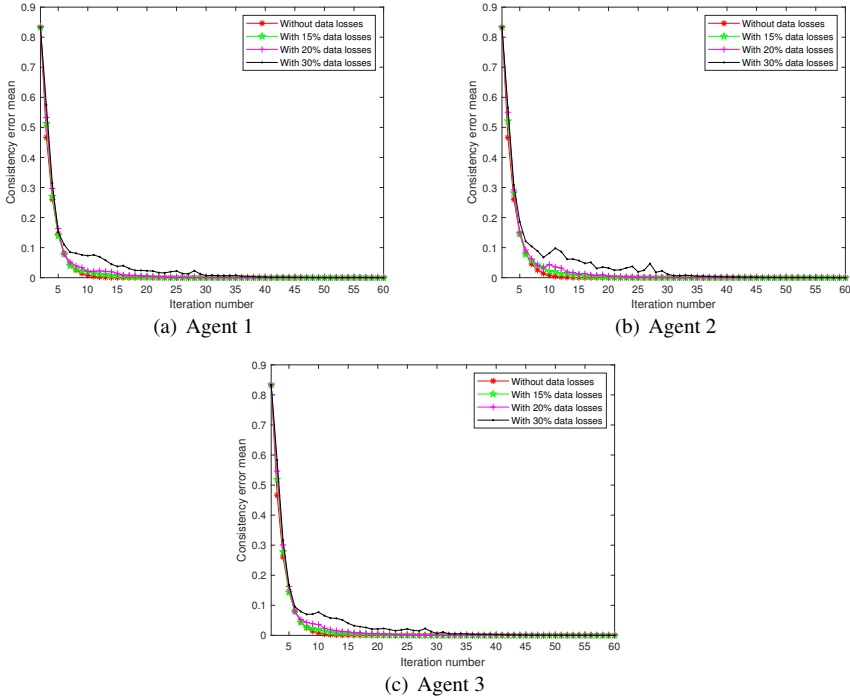


Figure 2. The convergence and stability of consistency error mean when $\Gamma = 0.6$

Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Conflict of interest

The authors declare no potential conflict of interests.

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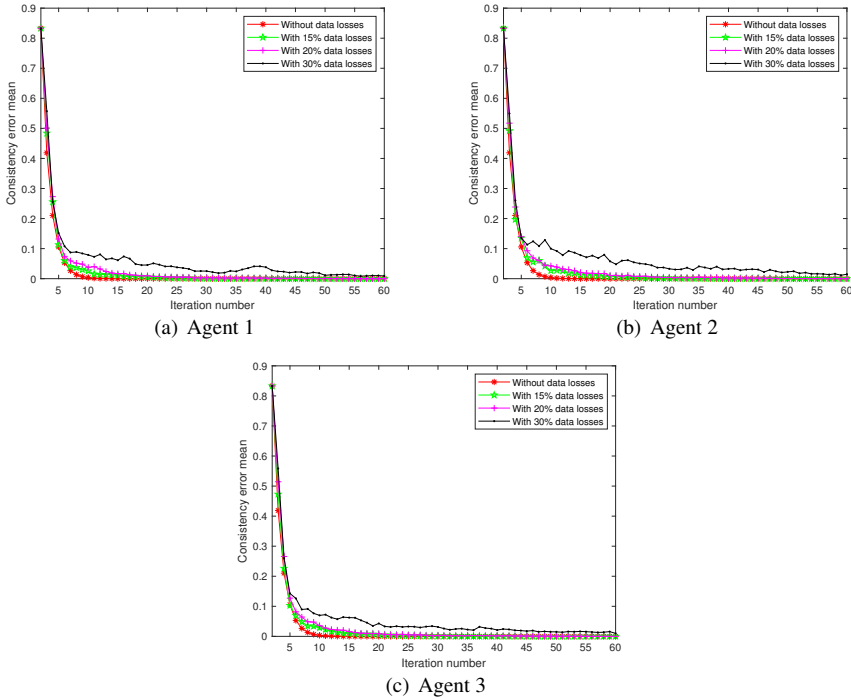


Figure 3. The convergence and stability of consistency error mean when $\Gamma = 0.7$

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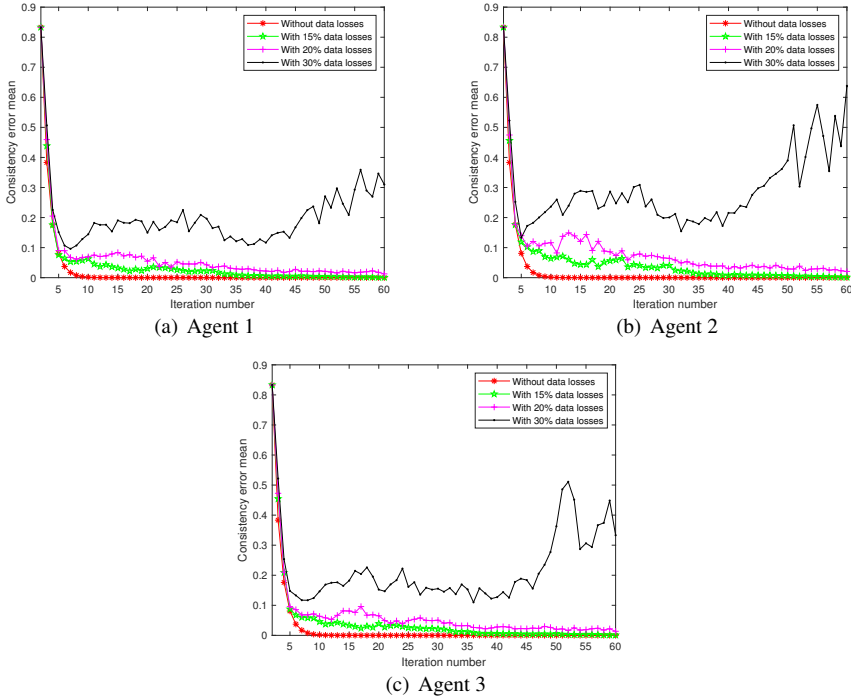


Figure 4. The convergence and stability of consistency error mean when $\Gamma = 0.76$

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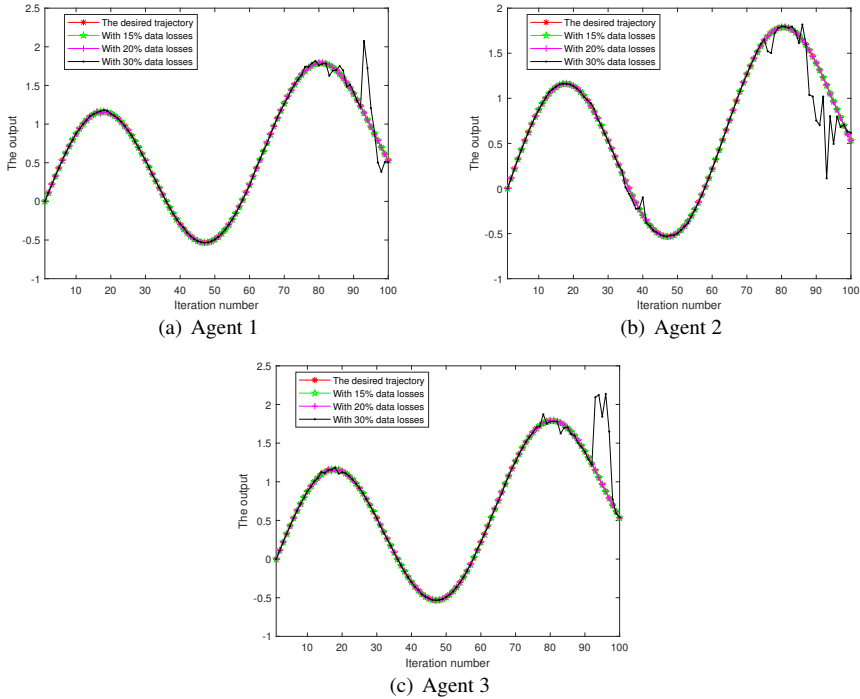


Figure 5. The output at 50-th iteration when $\Gamma = 0.7$

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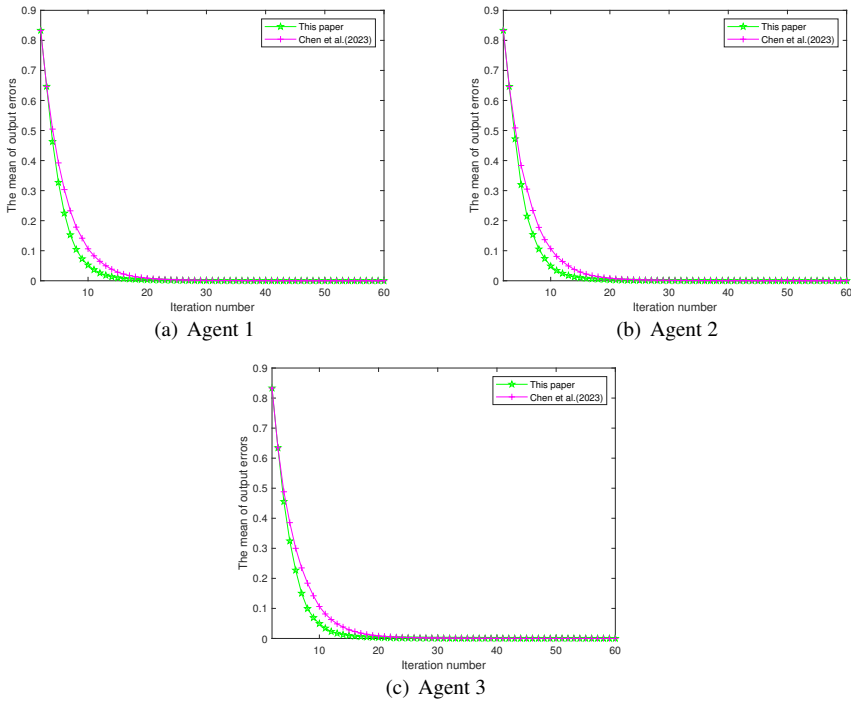


Figure 6. The convergence of the absolute mean of agent output errors using the proposed strategy or the strategy in [Chen et al. \(2023\)](#)

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